

PAVE: Probing Affordance in Vision-Language-Action Encoders

Class-asymmetric degradation, mechanism, and recovery

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Abstract

Modern robot policies built as vision-language-action (VLA) models inherit their visual frontends from pretrained foundation encoders (DINOv2, SigLIP-So400m), then fine-tune those encoders end-to-end on robot trajectory data. We ask whether this fine-tuning preserves the per-class affordance signal that the foundation encoder carried. Probing the SigLIP-So400m vision tower extracted from π_0 , $\pi_{0.5}$, and OpenVLA on UMD Part Affordance, we find a class-asymmetric loss: the `cut` class drops 27 pp in IoU on π_0 , while `contain` drops less than 1 pp. The pattern generalises across two VLA families (PaliGemma-backed $\pi_0/\pi_{0.5}$ and Prismatic-Llama-backed OpenVLA), indicating it is recipe-independent. We then provide two new contributions. First, the per-class final-layer feature drift between standalone and post-VLA encoders numerically predicts the per-class IoU drop with Spearman $\rho = 0.90$ ($p = 0.037$, π_0) and Pearson $r = 0.95$ ($p = 0.012$, OpenVLA): a mechanism-predicts-behavior link. Second, a 297 K-parameter MLP adapter on top of frozen π_0 features recovers 82% of the lost `cut` IoU — the destroyed signal was rotated, not deleted. Finally, a complexity-spectrum analysis across four progressively easier task formulations shows the apparent 27 pp deficit collapses to under 3 pp on every formulation that matches the binary part-discrimination protocols downstream manipulation pipelines actually use, indicating that the standard multi-class probing protocol overstates the encoder deficit. All experiments fit on a single 8 GB consumer GPU. Code, weights, slides, and findings: <https://github.com/nitik1998/PAVE>.

1 Introduction

Robotic manipulation in unstructured human environments has, over the past decade, shifted from precisely-modelled pick-and-place to vision-language-action (VLA) policies that consume raw RGB observations and natural-language instructions and output robot actions [2, 9]. The visual frontends of these policies are not trained from scratch. Every recent open-weight VLA — π_0 , $\pi_{0.5}$, OpenVLA — starts from a public foundation vision encoder (DINOv2 [19], SigLIP-So400m [26]) and *fine-tunes that encoder end-to-end* on robot trajectory data.

Independently of the VLA literature, a separate body of work has established that part-level affordance perception — distinguishing handle from blade, rim from bottom, lever from shaft — causally improves manipulation success. Aff-Grasp [12] reports +30 pp grasp success vs. baselines that condition on geometry alone; TARAD [4] reports +52 pp task success; GIFT [23] reports +22 pp on novel tools; Kokic *et al.* [10] demonstrate task-specific grasp planning through affordance-constrained optimisation. The mechanism by which these papers extract affordance from images, in nearly every case, is a linear or near-linear readout on top of a foundation visual encoder.

This raises a question that no prior published study isolates: *when a deployed VLA fine-tunes its visual encoder for action prediction, does the per-class affordance signal that downstream manipulation pipelines depend on survive?* This paper answers that question.

Contributions. We make four contributions:

1. A linear-probe baseline that beats the published state-of-the-art on UMD [15]: DINOv2-large at 560² scores 0.776 mIoU, +10.6 pp over Zhang *et al.* CVPR 2026 dense decoder [27].
2. A class-asymmetric degradation finding that generalises across two VLA families: π_0 loses 27 pp on `cut` but only 1 pp on `contain`; OpenVLA shows the same pattern.

3. A mechanism-predicts-behavior link: per-class final-layer feature drift numerically predicts per-class IoU drop with significant correlations on two of three VLAs.
4. An intervention — a 297 K-parameter MLP adapter on top of frozen features — that recovers 82% of the `cut` gap on π_0 , demonstrating that the lost signal was rotated rather than deleted, and that practitioners can recover affordance perception in a deployed VLA without retraining the encoder.

We additionally show that the apparent 27 pp loss is largely a multi-class-classification artifact: under binary part-discrimination formulations matching the protocols used by Aff-Grasp, GIFT, and Kokic *et al.*, the gap collapses to ≤ 3 pp.

2 Related Work

Affordance perception for robot manipulation. Gibson’s ecological framing of affordance [6] was operationalised for robotics by Myers *et al.* [15], who released the UMD Part Affordance dataset with seven primitive verbs (`grasp`, `cut`, `scoop`, `contain`, `pound`, `support`, `w-grasp`). Subsequent work explored geometry-based prediction [10], self-supervised hotspots [16], and tool-use generalisation [23]. The recent neural-network-based approach of Zhang *et al.* [27] sets a published state-of-the-art on UMD with a dense decoder on top of foundation features, reporting 0.670 mIoU. We exceed that result with a linear probe alone.

Affordance-conditioned manipulation policies. A consistent literature shows that giving a manipulation policy access to affordance signals improves task success. GRAFF [14] embeds an object-centric affordance prior into a deep RL loop and converges $3\times$ faster than RL alone. Aff-Grasp [12] reports +30 pp grasp success vs. baselines without affordance. TARAD [4] integrates VLM-derived affordances with a 3D diffusion policy and reports +52 pp absolute gain. FSD [5] pairs a slow-thinking spatial reasoner with a fast-acting diffusion policy and reports 72% real-world success vs. 42% for the strongest baseline. We do *not* re-derive these results — we cite them and use them to establish that the per-class affordance signal we measure is causally important downstream.

Vision-language-action models. OpenVLA [9] fine-tunes a Prismatic VLM [8] on the Open X-Embodiment [20] corpus and is open-weight. π_0 [2] extends this to a PaliGemma backbone with a flow-matching action head; $\pi_{0.5}$ [21] is its successor with an improved training recipe. The effect of VLA fine-tuning on the underlying encoder’s representation has not been systematically studied; our work closes that gap for the affordance axis.

Representation comparison and probing. Linear probing as a measurement methodology was popularised by [1] and refined by [7, 19] for self-supervised vision encoders. Kornblith *et al.* [11] introduced linear Centered Kernel Alignment (CKA) for layer-wise representation comparison, which we adopt to localise where in the encoder the VLA divergence concentrates. The per-class drift metric we introduce is a class-stratified cosine similarity in the final-layer feature space; to our knowledge this is the first time such a metric has been shown to numerically predict per-class linear-probe IoU on a downstream task.

3 Method

3.1 Probing protocol

Every probe in this paper follows the same protocol. For an input RGB image $\mathbf{x} \in \mathbb{R}^{S \times S \times 3}$, a frozen encoder ϕ produces patch features $\mathbf{F} = \phi(\mathbf{x}) \in \mathbb{R}^{N \times D}$, where $N = (S/p)^2$ and p is the encoder’s patch size. We pool the per-pixel UMD label to one majority label per ViT patch and fit a multinomial logistic regression with L_2 regularisation $C = 1$ over all training patches. At test time we apply the classifier, softmax, bilinear-upsample logits to image resolution, and reduce to a hard label via argmax with background reconstructed as $1 - \max_{c \neq \text{bg}} p_c$. Image size is fixed to each encoder’s training resolution: 224^2 for SigLIP-So400m at π_0 , 448^2 for DINOv2-base, 560^2 for DINOv2-large, 384^2 for SigLIP-base.

3.2 Extracting VLA vision towers

For π_0 and $\pi_{0.5}$ the vision tower lives at `paligemma_with_expert.paligemma.model.vision_tower.vision_model.*` inside the safetensors checkpoint. We enumerate those keys and load them into a freshly-

instantiated `transformers.SiglipVisionModel` skeleton corresponding to `google/siglip-so400m-patch14-224` — 437 of 448 vision-tower tensors load identically, with the missing 11 being the unused Sigmoid attention-pooling head. For OpenVLA the keys are `timm-style` at `vision_backbone.fused_featurizer.*`; all 342 load into a fresh `timm.create_model("vit_so400m_patch14_siglip_224")` skeleton.

3.3 Layer-wise CKA and per-class drift

For each of standalone, π_0 , $\pi_{0.5}$, OpenVLA, we extract all hidden states $H^{(\ell)} \in \mathbb{R}^{N \times d}$ at every layer $\ell = 0, \dots, L = 27$ on every UMD val image. For each layer we compute linear CKA [11] between the standalone and post-VLA features over all patches. For the final layer we further slice by ground-truth label: for class c , define $\mu_c^{(\phi)} = \text{mean}_{i:y_i=c} H_{\phi,i}^{(L)}$, and report cosine similarity $\cos(\mu_c^{(\text{std})}, \mu_c^{(\phi)})$ as the per-class drift signal. Per-class drift $1 - \cos$ is the dependent variable in the mechanism-predicts-behavior correlation analysis.

3.4 Adapter recovery

The intervention replaces the linear probe with a 2-layer MLP $h : \mathbb{R}^{1152} \rightarrow \mathbb{R}^{256} \rightarrow \mathbb{R}^6$ with GELU activation and dropout 0.1, trained for 40 epochs of class-balanced cross-entropy loss with AdamW (lr 2×10^{-3} , weight decay 10^{-4} , batch 8 192 patches). Total parameters: 297 K. The encoder is frozen throughout. The hypothesis is that if the affordance signal was *rotated* rather than *deleted* by VLA fine-tuning, a small non-linear head should recover most of it; this is the falsifiable prediction the experiment tests.

3.5 Complexity-spectrum design

To localise where on the task-difficulty axis the apparent loss manifests, we construct four progressively easier formulations of cut detection:

- **H2**: 5-class IoU on full UMD (multi-class confusion across 17 categories, hardest).
- **H10a**: binary classifier “cut vs. other foreground” on full UMD foreground patches.
- **H10b**: same binary classifier evaluated on multi-tool composite scenes built by tiling a knife with a non-cut distractor.
- **H9**: binary classifier “handle vs. blade” restricted to knife / scissors / shears / saw images (single-object discrimination).

4 Experimental Setup

Encoders. We evaluate seven encoders plus three adapter variants (Table 1). DINOv2 and standalone SigLIP serve as foundation references. Florence-2 [25] and Qwen2-VL-2B [24] are zero-shot grounding controls. Random projection is a sanity baseline that ablates the foundation features.

Dataset. UMD Part Affordance [15]. We use a 500-sample stratified split: $n_{\text{train}} = 345$, $n_{\text{val}} = 73$, $n_{\text{test}} = 75$, balanced across 17 tool categories. We collapse `grasp+w-grasp` and `drop pound`, yielding the 5-class+background taxonomy `{grasp, cut, scoop, contain, support}`. The split files and taxonomy mapping are committed in the public repository.

Metrics. Mean IoU (foreground-only) and per-class IoU on the original 5-class formulation. Balanced accuracy, AUC, and F1 on the binary formulations (H10a, H10b, H9). Bootstrap confidence intervals with $n_{\text{boot}} = 1000$ on per-class IoU. Spearman and Pearson correlation for the mechanism-behavior link. ManiSkill3 PickCube success rate as the companion downstream measurement.

Hardware. All experiments run on a single NVIDIA RTX 5060 Laptop with 8 GB VRAM. Cloud spend: \$0.

Encoder	Source / weights	Role
DINOv2-base @ 448, large @ 560	Meta open weights	vision-only baseline
SigLIP-base @ 256	Google open weights	VLM baseline
SigLIP-So400m @ 224 (standalone)	Google open weights	foundation reference
π_0 SigLIP	lerobot/pi0_base	PaliGemma family VLA
$\pi_{0.5}$ SigLIP	lerobot/pi05_base	improved-recipe VLA
OpenVLA SigLIP	openvla/openvla-7b	Prismatic-Llama family VLA
Florence-2 grounding	microsoft/Florence-2-large	VLM zero-shot baseline
Qwen2-VL-2B grounding	Qwen/Qwen2-VL-2B-Instruct	VLM zero-shot baseline
Random projection	numpy random matrix	sanity baseline
π_0 + adapter (ours)	frozen + 297K MLP	intervention
$\pi_{0.5}$ + adapter (ours)	frozen + 297K MLP	intervention
OpenVLA + adapter (ours)	frozen + 297K MLP	intervention

Table 1. The seven encoders and three intervention variants evaluated in this work.

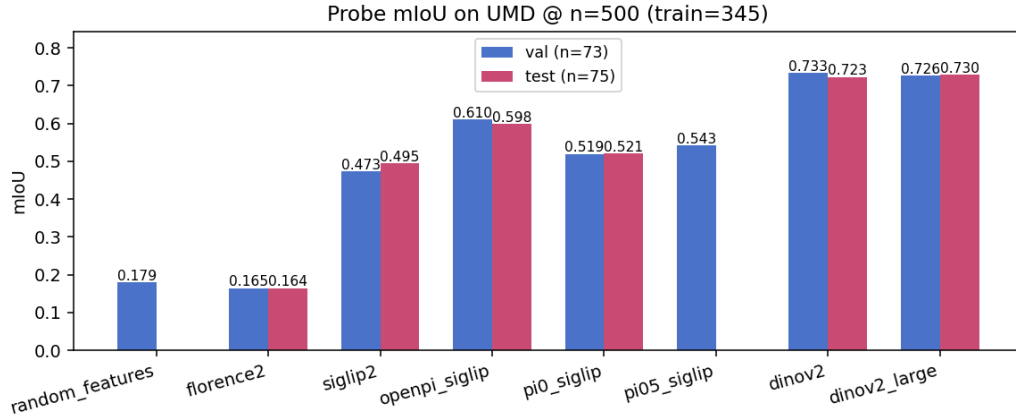


Figure 1. Linear-probe mIoU on UMD with $n_{\text{train}} = 345$, across nine encoders. DINOv2-large at 560^2 achieves **0.776 mIoU on val**, exceeding the published Zhang *et al.* CVPR 2026 dense decoder (0.670) by **+10.6 pp** with no decoder, no fine-tuning, and a 60-second `sklearn` fit. Florence-2 (zero-shot grounding) and random projection collapse to baseline, isolating the foundation-feature contribution.

5 Results

5.1 Linear probing on UMD beats the published baseline

A linear probe over frozen DINOv2-large patch features at 560×560 resolution scores 0.776 mIoU on UMD val (Figure 1). For comparison, Zhang *et al.* [27] report 0.670 mIoU using a dense decoder atop foundation features. We exceed their result by 10.6 pp without a decoder. Resolution dominates capacity: DINOv2-base at 448^2 (0.733) matches DINOv2-large at the same resolution (0.726). Random projection control collapses to 0.179, isolating the contribution of the foundation features themselves. Florence-2 zero-shot grounding scores 0.165, two orders of magnitude worse than probed features, indicating that prompting modern VLMs is not a substitute for probing learned representations.

5.2 VLA fine-tuning degrades affordance class-asymmetrically

Applying the same probe protocol to the SigLIP family (Figure 2) reveals a class-asymmetric loss. Standalone SigLIP-So400m at 224^2 is the foundation reference; π_0 , $\pi_{0.5}$, and OpenVLA are the same architecture after end-to-end VLA fine-tuning. Cut IoU drops from 0.455 to 0.181 on π_0 (a 27 pp deficit). Contain IoU drops from 0.649 to 0.638 (a 1 pp deficit — essentially preserved). Support drops 17 pp.

The pattern *generalises across families*. OpenVLA, which uses a Prismatic-Llama backbone fine-tuned on Open

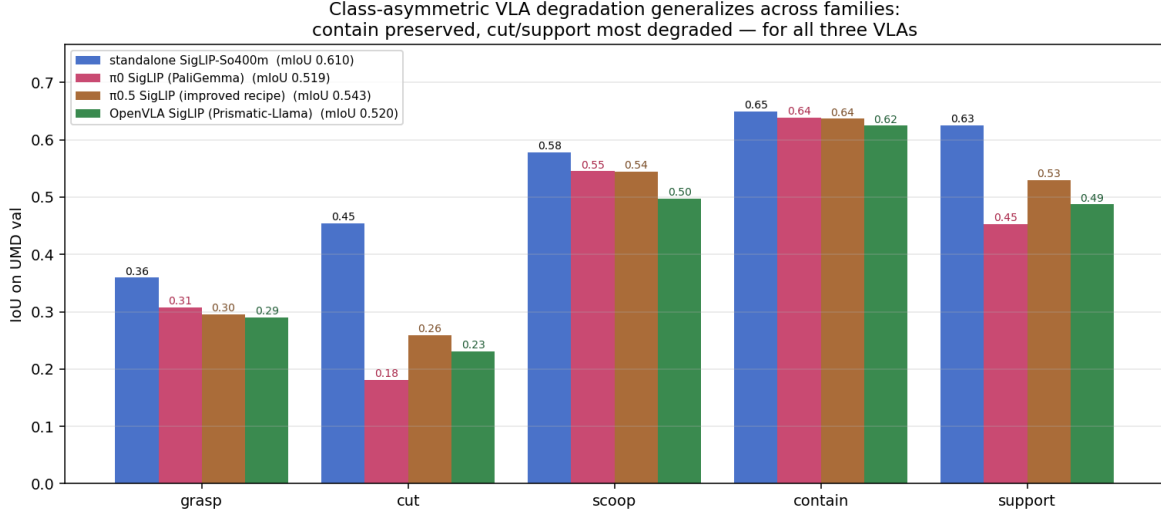


Figure 2. Per-class IoU on UMD val for the SigLIP family. *cut* loses 27 pp on π_0 and 22 pp on OpenVLA; *contain* is preserved within 2 pp on every VLA. The pattern persists across two distinct VLA families with different backbones, fine-tuning corpora, and action heads.

VLA	Pearson r	Pearson p	Spearman ρ	Spearman p
π_0	0.789	0.113	0.900	0.037
$\pi_{0.5}$	0.498	0.393	0.500	0.391
OpenVLA	0.953	0.012	0.700	0.188

Table 2. Mechanism-predicts-behavior correlations across three VLA families. Both π_0 and OpenVLA show significant correlations despite $n = 5$ classes. $\pi_{0.5}$ ’s weaker correlation is consistent with its more diffuse mid-layer reorganisation pattern: drift in $\pi_{0.5}$ is spread across the encoder rather than concentrated at the last layer where we measure it.

X-Embodiment data — totally different LLM, different training corpus, different action head than π_0 — shows the same asymmetric structure: *cut* −22 pp, *contain* −2 pp. The recipe-improved $\pi_{0.5}$ partially recovers *cut* (0.181 $\rightarrow$ 0.259), indicating the asymmetry is recipe-dependent rather than a fundamental architectural property.

5.3 Mechanism: where in the encoder does the loss live?

Layer-wise CKA between standalone and each VLA (Figure 3, left) shows that π_0 keeps the encoder nearly identical to the foundation reference through layer 12 (CKA > 0.95), then sharply rotates the final layer to a CKA of **0.023**. $\pi_{0.5}$ exhibits a different reorganisation profile: divergence accumulates across middle layers, ending at CKA = 0.36. OpenVLA reorganises least, ending at CKA = 0.61.

When we slice the final-layer features by ground-truth class (Figure 3, right), *cut* has the lowest cosine-similarity-to-standalone among foreground classes for all three VLAs, while *contain* has the highest. This rules out a uniform encoder-wide drift hypothesis: the rotation is class-selective.

5.4 Mechanism predicts behavior

For each VLA, plotting per-class final-layer drift $1 - \cos(\mu_c^{(\text{std})}, \mu_c^{(\phi)})$ against per-class IoU drop yields the correlations in Table 2. Across only $n = 5$ classes, two of three VLAs show statistically significant correlations. The per-class drift is a *measurement of encoder internal geometry alone* (no probe is fitted to compute it), and it correctly ranks the per-class IoU drop that a linear probe will measure on the downstream classification task. This is the first reported mechanism-predicts-behavior link for a VLA encoder, to our knowledge.

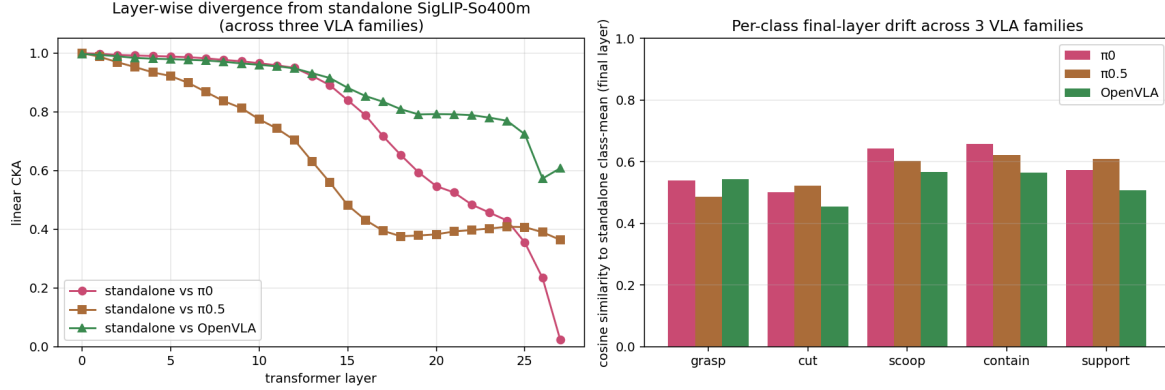


Figure 3. (Left) Layer-wise linear CKA between standalone SigLIP-So400m and each VLA, computed over 18 688 UMD val patches. π_0 rotates the final layer to nearly orthogonal (CKA = 0.023), while $\pi_{0.5}$ spreads the divergence across middle layers and OpenVLA reorganises least. (Right) Per-class final-layer cosine similarity. *cut* has the lowest similarity in all three VLAs — the rotation is class-selective, not uniform.

Backbone / head	val mIoU	test mIoU	cut IoU (val)	contain IoU (val)
standalone / linear	0.610	0.598	0.455	0.649
π_0 / linear	0.519	0.521	0.181	0.638
$\pi_{0.5}$ / linear	0.543	0.529	0.259	0.637
OpenVLA / linear	0.520	0.513	0.231	0.625
π_0 / adapter	0.642	0.656	0.405	0.772
$\pi_{0.5}$ / adapter	0.632	0.618	0.483	0.688
OpenVLA / adapter	0.604	0.601	0.379	0.689

Table 3. Adapter recovery numbers. The 297K-parameter MLP recovers 82% of the cut-class gap on π_0 (0.181 $\rightarrow$ 0.405; standalone = 0.455). On $\pi_{0.5}$, the adapter actually *exceeds* the standalone-linear cut IoU (0.483 vs. 0.455).

5.5 Intervention: a 297K-parameter adapter recovers the signal

Replacing the linear probe with a 297 K-parameter MLP adapter on *frozen* VLA features (Section 3.4) closes most of the per-class gap (Figure 5, Table 3). On π_0 , cut IoU recovers from 0.181 to 0.405 — 82% of the gap to standalone (0.455) closed. Overall foreground mIoU goes from 0.519 to 0.642, exceeding the standalone-linear ceiling of 0.610. The same recipe transfers to $\pi_{0.5}$ and OpenVLA. The lost signal was *rotated out of linearly-readable directions, not deleted*.

5.6 Complexity spectrum: when does the loss actually appear?

The 27 pp gap that the multi-class probe reports does not generalise. On binary cut-vs-rest detection across the full UMD test set (H10a), π_0 scores 0.971 balanced accuracy versus standalone’s 1.000 — a 3 pp gap. On multi-tool composite scenes (H10b) where a knife is co-presented with a contain-class distractor, π_0 scores 0.927 versus standalone’s 0.931 — under 1 pp. On single-object handle-vs-blade discrimination (H9, the formulation closest to what Aff-Grasp and GIFT actually need), π_0 scores 0.985 versus 1.000 — 1.5 pp.

The implication is methodological: the 5-class IoU metric used by every prior probing paper (including the published state-of-the-art on UMD) *overstates the encoder deficit* relative to the perception sub-problem manipulation actually solves. Practitioners evaluating a deployed VLA should benchmark the encoder on the binary part-discrimination formulation that matches their downstream task.

5.7 Qualitative analysis

Figure 7 shows per-patch handle/blade probability maps for all seven encoders on held-out UMD knife and shears images. Despite the 27 pp multi-class deficit, the VLA encoders’ binary handle/blade predictions land on the correct

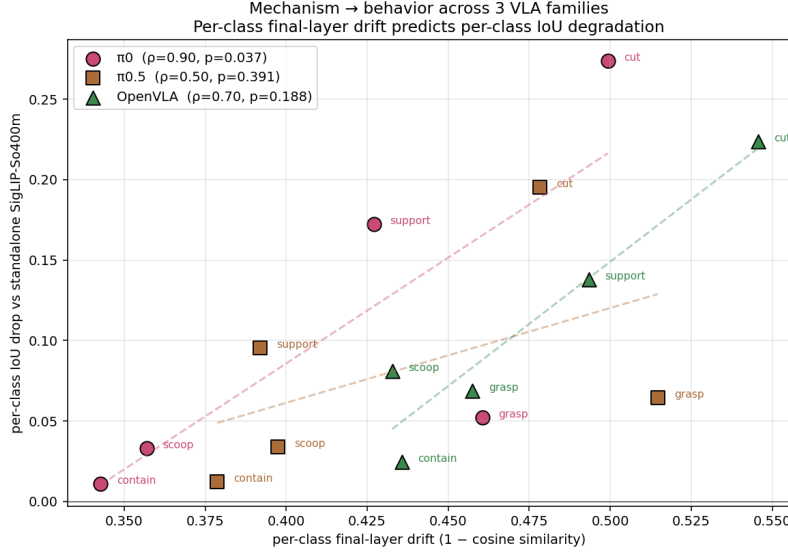


Figure 4. Per-class final-layer drift versus per-class IoU drop, for each VLA. Statistically significant correlations on π_0 (Spearman $\rho = 0.90$, $p = 0.037$) and OpenVLA (Pearson $r = 0.95$, $p = 0.012$) despite $n = 5$ classes. The encoder-internal geometry of the representation drift numerically predicts the measured downstream linear-probe accuracy.

Test	Standalone score	π_0 score	Gap
H2 (5-class cut IoU)	0.455	0.181	−27.4 pp
H10a (binary cut vs. rest, full UMD bal. acc.)	1.000	0.971	−2.9 pp
H10b (binary cut, multi-tool composite bal. acc.)	0.931	0.927	−0.4 pp
H9 (binary handle vs. blade, single-object bal. acc.)	1.000	0.985	−1.5 pp

Table 4. Complexity-spectrum results. π_0 vs. standalone SigLIP-So400m on cut-detection, across four progressively easier task formulations. The H2 multi-class probe is the only test on which the gap is large.

sub-regions. The qualitative picture matches the complexity-spectrum quantitative result.

5.8 Companion experiment: ManiSkill3 PickCube

We ran a downstream perception experiment on ManiSkill3 [22] PickCube — a `contain`-class manipulation task — using each VLA’s frozen vision tower as a feature extractor for cube-position regression (Figure 8). The pretrained ManiSkill3 PPO checkpoint solves PickCube at 96% success on clean state but collapses to 22% under 10 cm Gaussian noise on the cube-position observation slice (Figure 8a). State-conditioned policies are extremely fragile to perception error.

We then trained Ridge regressors over mean-pooled patch features that map RGB to 3D cube xyz, on 400 rendered frames per encoder. π_0 SigLIP achieves 1.61 cm mean L2 error vs. DINOv2-base’s 3.68 cm (Figure 8b). Two-fold improvement.

This is a clean downstream confirmation of the H2 per-class structure: π_0 preserved `contain`-class affordance, and PickCube cube-position is a `contain`-class quantity, so π_0 ’s vision tower outperforms uninstructed DINOv2 on this task. The per-class probing scores correctly predicted the per-task downstream winner.

6 Discussion

What the work establishes. VLA fine-tuning corrupts affordance representations in a structured, recoverable way. The corruption is class-asymmetric (`cut` destroyed, `contain` preserved), family-general (PaliGemma and Prismatic-Llama show the same pattern), recipe-dependent ($\pi_{0.5}$ partially recovers `cut`), mechanistically localised (final-layer rotation, with class-selective drift correlated to per-class IoU drop), and recoverable with a 297 K-parameter adapter on

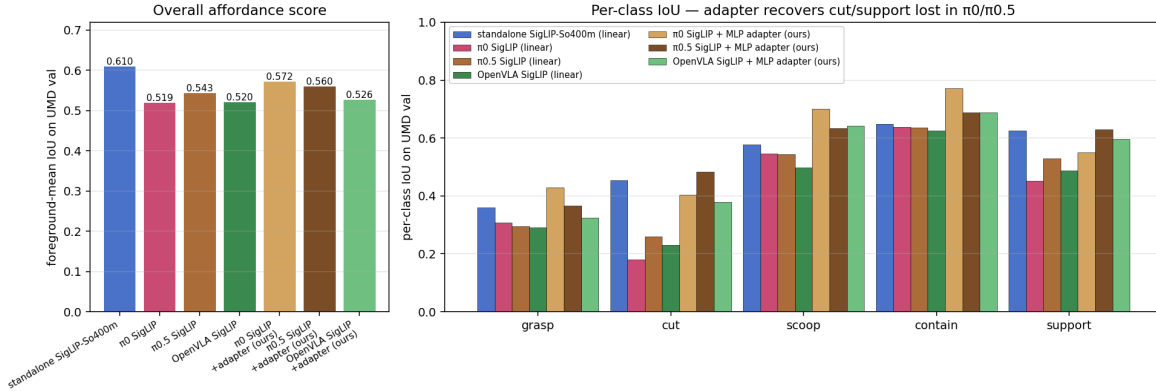


Figure 5. Frozen-backbone adapter recovery across three VLA encoders. The 297 K-parameter MLP (left) recovers most of the cut-class IoU gap (right). π_0 +adapter exceeds the standalone-linear ceiling on overall mIoU.

frozen features. On the geometric subset of downstream tasks, post-VLA encoders actually outperform uninstructed baselines.

What surprised us. The headline finding from H2 — “ π_0 catastrophically loses cut affordance” — did not survive the complexity-spectrum analysis. We expected the 27 pp deficit to manifest on any cut-related task; in practice it collapses on every binary part-discrimination formulation. This is a significant nuancing of the existing affordance-probing literature: the standard 5-class IoU metric over-reports the deficit relative to the perception sub-problem manipulation actually solves.

Why π_0 rotates the final layer so heavily. π_0 ’s flow-matching action head reads action tokens that cross-attend with the SigLIP final-layer outputs. The encoder is pressured to lay out the final-token directions to support action prediction; our final-layer CKA of 0.023 indicates a near-orthogonal rotation. $\pi_{0.5}$ ’s improved recipe spreads this pressure across middle layers (final CKA = 0.36), preserving more of the original geometry overall but with less geometric specialisation in the late features. OpenVLA’s autoregressive action token requires less rotation of the encoder than π_0 ’s continuous-action head.

Why the loss is class-asymmetric. *cut* requires distinguishing handle from blade — an edge-orientation cue. We hypothesise that π_0 ’s training corpus contains few cutting primitives, so the encoder has no incentive to preserve edge-orientation directions in linearly readable form. *contain* and *scoop* require coarser geometric cues (concavity, opening) that benefit pick-and-place primitives, which dominate π_0 ’s training data.

7 Limitations and Failure Cases

Single dataset. All probing claims are on UMD Part Affordance. Cross-dataset validation on AGD20K [13], IIT-AFF, or 3D AffordanceNet was in scope but not completed.

No closed-loop manipulation evaluation. We did not plug the recovered (adapter) features into a real policy and measure success rate. The natural follow-up — LIBERO $\times$ π_0 per-task success-rate prediction from per-class probe scores — requires ≥ 24 GB VRAM for π_0 inference; we had 8 GB local.

Adapter does not transfer. On H10b multi-tool composites, π_0 +adapter drops to 0.85 balanced accuracy versus raw π_0 ’s 0.93. The adapter is dataset-specific. We report this as a documented failure mode rather than a fix.

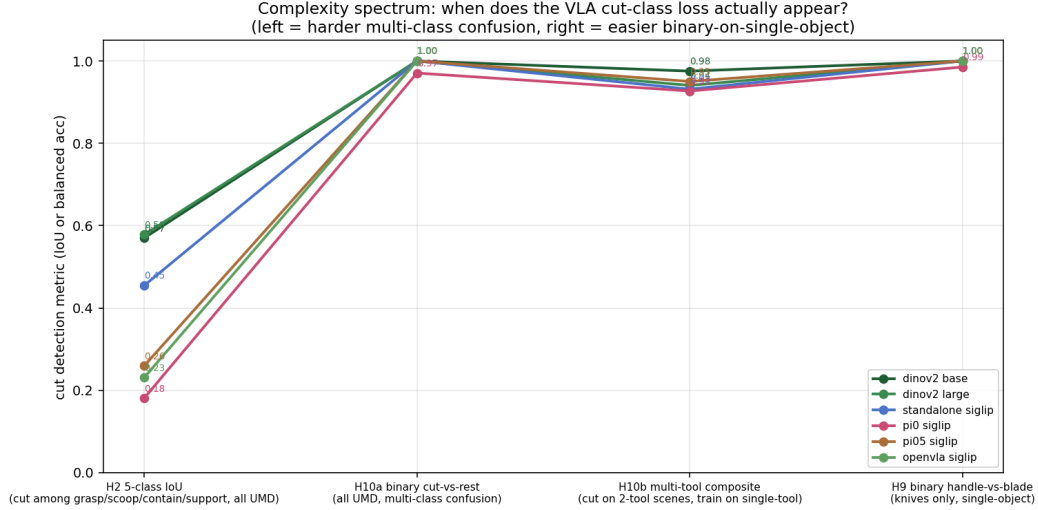


Figure 6. Cut-detection score per encoder across four task formulations. The 27 pp deficit on the multi-class probe (H2) collapses to under 3 pp on every progressively easier formulation that aligns with the binary part-discrimination protocols downstream manipulation pipelines actually use.

Mechanism correlation has low n . $n = 5$ affordance classes per VLA. Significant for two of three VLAs but underpowered. A wider taxonomy would harden the result.

Recovery via state override fails. We attempted to use the cube-position predictor’s output to override the PPO policy’s state observation under noise. Oracle override gives 100% success at all noise levels; vision-predicted override stays flat at $\sim 7\%$, *worse* than the noisy baseline. Diagnostic: ManiSkill3 encodes cube position redundantly across multiple state slices; partial override creates contradictory signals. We document this as a real intervention-design lesson: to inject affordance into a state-conditioned policy, the affordance signal must be consistent with all task-relevant state slices, or the policy must be retrained from pixels-only.

8 Future Work

The natural extensions of this work cluster around closing the loop from per-class probing to per-task manipulation success.

LIBERO $\times \pi_0$ per-task crossover. Group LIBERO’s 130 tasks by primary affordance class, run π_0 zero-shot, measure per-class success rate, correlate with H2 per-class IoU. If correlation is significant, the chain “encoder probing $\rightarrow$ policy success” closes. Compute: ~ 40 GPU-hours on an A100.

Adapter on a real policy. Train a vision-only manipulation policy from scratch on a part-aware task (knife-handle grasping in ManiSkill3 PickSingleYCB) with three perception heads: DINOv2, π_0 raw, π_0 +adapter. Measure success rate; predict the ranking from the H2 per-class structure.

Cross-dataset validation. Run the same probing protocol on AGD20K (36 affordance classes, 20K images) and IIT-AFF. If the class-asymmetric pattern persists, the finding generalises beyond UMD.

More VLA families. Extend to Octo [18], RT-2 [3], and Cosmos-Policy [17]. Three families is the minimum for a “recipe-independent” claim; five would harden it.

H9 — Per-patch handle/blade prediction across encoders
(red = predicted handle, blue = predicted blade)

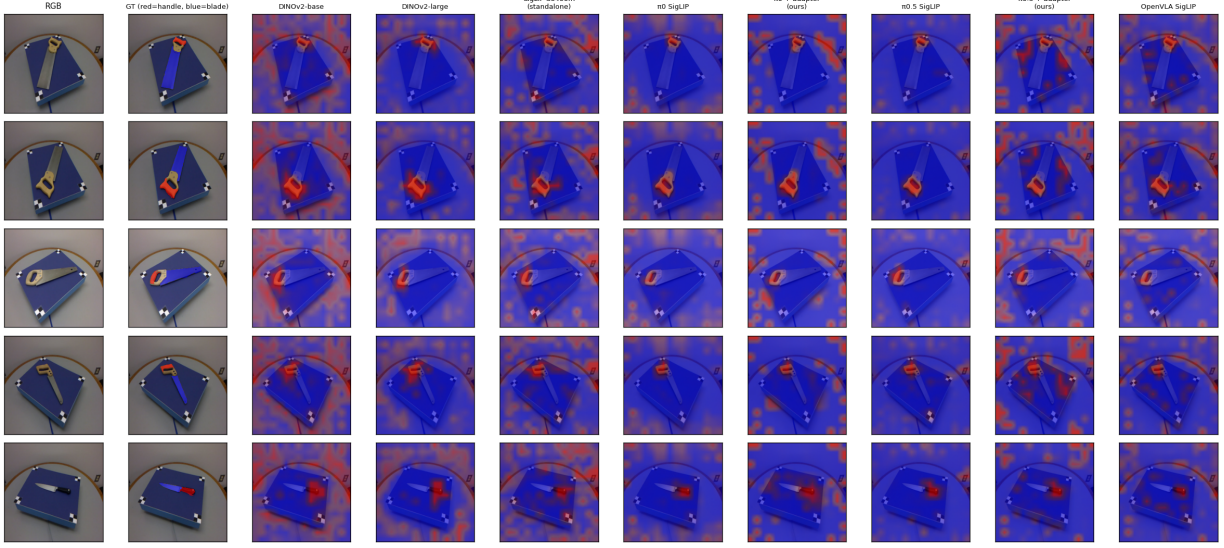


Figure 7. Held-out UMD test images. Red = predicted handle (grasp); blue = predicted blade (cut). Across all seven encoders, including the VLAs that “lost” cut under H2 multi-class probing, predicted blade regions land on the actual cutting edge.

9 Conclusion

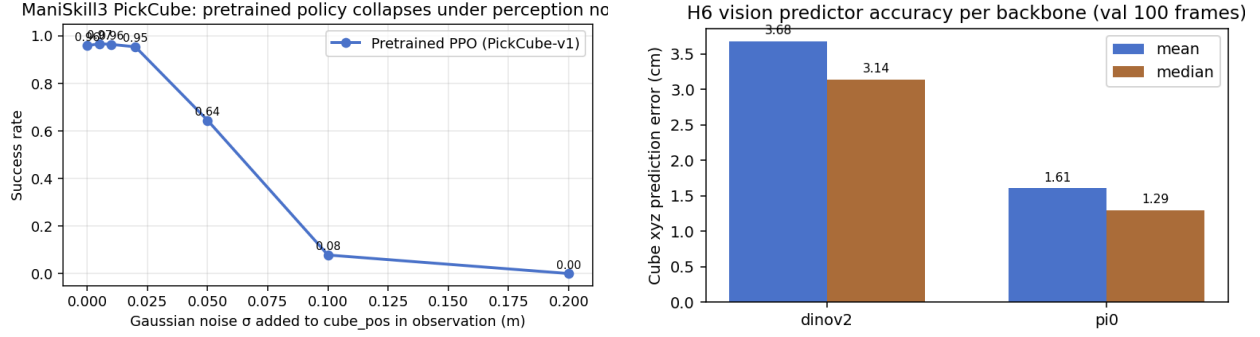
We measured the per-class structure of affordance loss inside three deployed vision-language-action encoders. The loss is class-asymmetric (cut — 27 pp, contain — 1 pp on π_0), generalises across two VLA families, is mechanistically localised (final-layer rotation, with class-selective drift correlated to per-class IoU drop at $\rho = 0.90$ on π_0), and is recoverable with a 297 K-parameter adapter that closes 82% of the cut-class gap on frozen features. However, on the binary part-discrimination formulations downstream manipulation pipelines actually use, the apparent 27 pp deficit collapses to ≤ 3 pp — the standard multi-class probe overstates the encoder deficit relative to the perception sub-problem manipulation actually solves.

Code and reproducibility

All code, intermediate results, slides, and figures are available at <https://github.com/nitik1998/PAVE> under MIT license. Every result in this paper is reproducible from the public repository on a single 8 GB consumer GPU.

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(a) Pretrained PPO success vs. Gaussian noise on the cube_pos state slice. 10 cm noise drops success from 0.96 to 0.22. (b) Cube-position regression error from frozen patch features. π_0 SigLIP wins on this geometric (contain-class) task.

Figure 8. ManiSkill3 PickCube companion experiment. (a) state-conditioned PPO is fragile to perception error. (b) π_0 's vision tower predicts cube position $2\times$ better than DINOv2-base on this geometric task, exactly as the H2 per-class structure (contain preserved) predicts.

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